An Internship Report

Of

Software Development Internship at W3villa Technologies

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)

By

Roll Number of Student: 2410030486

Name of Student: ARPIT SRIVASTAVA

Batch: 2024-2028

CANDIDATE'S DECLARATION

I**, ARPIT SRIVASTAVA,** do hereby declare that the internship report titled **“Software Development Internship at W3villa Technologies”** is an authentic record of the work, learning, and technical experience gained by me during my internship tenure. The content presented in this report is based on my own tasks, assignments, and practical project implementations carried out during the internship.

I affirm that all references and tools used have been duly acknowledged and that this report has not been submitted to any other university or institution for the award of any degree or diploma. I submit this report in partial fulfilment of the requirements for the degree of **Bachelor of Technology in Computer Science and Engineering** at the **School of Computer Science and Engineering, IILM University, Greater Noida.**

Signature: ___________________

Student Name: ARPIT SRIVASTAVA

Roll No.: 2410030486

ACKNOWLEDGMENT

I take this opportunity to express my heartfelt gratitude to everyone who supported, guided, and encouraged me throughout my internship program at **W3villa Technologies Private Limited**. This experience gave me an opportunity to apply my classroom knowledge to practical tasks and understand how software development work is carried out in a professional environment.

I am deeply thankful to the technical leads, mentors, and the entire engineering team at W3villa Technologies for their insightful guidance and constructive feedback. Working on real-world assignments—ranging from interpreting Functional Requirement Documents (FRDs) and scaling a Task Management platform into a feature-rich SaaS system (ORBIQ OS), to performing computer vision video annotations using CVAT—has been immensely educational. These tasks instilled in me the core principles of software engineering: attention to detail, debugging patience, and adaptability to evolving product requirements.

I extend my sincere thanks to **IILM University, Greater Noida**, and the faculty members of the **School of Computer Science and Engineering** for their constant encouragement and for providing the academic framework that facilitated this internship opportunity.

Finally, I express my deepest gratitude to my parents and family for their unwavering blessings, faith, and support throughout this learning journey.

Signature: ___________________

Student Name: ARPIT SRIVASTAVA

Roll No.: 2410030486

TABLE OF CONTENTS

3. INTERNSHIP COMPLETION CERTIFICATE

W3villa technologies

contact@w3villa.com
B-49, B Block, Sector 59,
Noida, Uttar Pradesh 201301

Ref: WT/Intern/IL/072026/001 **Date:** 31 July 2026

Arpit Srivastava
Intern-Software Developer
Noida

Subject: Internship Completion Letter

Respected Sir/ Madam,

This is to inform you that **Arpit Srivastava** has completed his internship as an **Intern – Software Developer** from our company, **W3villa Technologies Private Limited**. The internship period was from **08 June 2026** & completed on **31 July 2026**.

He is hard-working and dedicated to the work and holds the good potential to learn new things. He has done an amazing job during the internship and we hereby confirm that the above-mentioned information is true and does not hold any false information.

Sincerely,

For W3villa Technologies Private Limited

For W3VILLA TECHNOLOGIES PVT. LTD.
Shweta Malaviya
Human Resource

Sincerely,
Shweta Malaviya
(Head- Human Resources)

CIN: U74110DL2014PTC273849
Registered office: Innov8 CP2 44, Backary Portion, 2nd Floor, Regal Building, Connaught Place, Delhi, India, 110001
www.w3villa.com

INTERNSHIP OFFER LETTER

Gmail **Arpit Srivastava <arpitsrivastava1101@gmail.com>**

W3villa Technologies Pvt. Ltd. || Internship Offer – Software Developer Intern

Aditi Srivastava <aditi.srivastava@w3villa.com> 4 June 2026 at 16:31
To: Arpit Srivastava <arpitsrivastava1101@gmail.com>
Cc: HR Group <hr@w3villa.com>

Dear Arpit,

Greetings from **W3villa Technologies!**

We are pleased to let you know that we intend to hire you for the position of **Software Developer Intern (*Unpaid*).** Please let me know the starting date of employment that shall not be later than **8 June 2026.**

Reporting time : 11:00 AM

Location : B-49, B Block, Sector 59, Noida, Uttar Pradesh 201301

Contact Person : Ms. Aditi Srivastava

Please **confirm your acceptance of the offer of Internship** by replying to this email, ASAP**.** Also, bring the photocopy of the below-mentioned documents on your joining day:

- Xth Certificate & mark sheets
- XIIth Certificate & mark sheets
- Degree Certificate & Semester/year-wise mark sheets
- Master's Certificate & Semester/year-wise mark sheets (if any)
- Aadhar & Pan copy
- NOC or LOR from the college/University
- Three passport-sized color photographs.

You are required to bring your laptop throught the internship duration.

Note: <u>This offer is valid for 8 June 2026 only. If you are failing to join us on the expected date, this opportunity will be invalid and considered as oblivion and unoccupied.</u>

To accept the company's offer, kindly confirm with your acknowledgement and documents.

If you have any further queries or any information required, please reach us at @Aditi Srivastava

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Aditi Srivastava
Executive- Human Resources

W3villa Technologies Private Limited
B-49, Sector - 59
Noida, U.P - 201301
Phone: +91-9319756176
Email: aditi.srivastava@w3villa.com

4. PROJECT DESCRIPTION

4.1 INTRODUCTION

This report details my experience during my software engineering internship at **W3villa Technologies Private Limited, Noida**, from **08 June to 31 July 2026**. Fulfilling the degree requirements for my **B.Tech in Computer Science and Engineering at IILM University, Greater Noida**, the internship gave me a realistic look into how software is developed and maintained outside a classroom environment.

My core workload was split across **full-stack development and computer vision data processing**. The major coding focus revolved around **TaskFlow**, a task management project developed from the initial **Functional Requirement Document (FRD)**, which I later expanded and evolved into **ORBIQ**, a productivity-focused SaaS workspace built on the **MERN stack (MongoDB, Express.js, React, Node.js)**. Starting from the initial requirements, I worked on frontend UI components, backend REST API integration, application features, and debugging during development and deployment. Dealing with unexpected issues, API integration problems, and changes in requirements gave me a clearer picture of the iterative nature of web application development.

Parallel to development, I spent time preparing computer vision datasets in **CVAT (Computer Vision Annotation Tool)**. My work involved annotating video data by drawing **bounding boxes and polygon annotations**. The completed work was reviewed, and when mistakes were identified, I corrected them and resubmitted the annotations. Although this work was quite different from writing code, it showed me the importance of accuracy and consistency when preparing data for machine learning and computer vision applications.

Balancing these two areas gave me exposure to several stages of the development process, including **requirement analysis, database work, frontend and backend integration, dataset verification, and deployment debugging**.

4.2 Organization Profile

W3villa Technologies is a software development company based in Noida, Uttar Pradesh, providing technology solutions and software development services to businesses across different industries. The company works on a range of digital solutions including web application development, mobile application development, e-commerce solutions, custom software development, blockchain, and other technology-driven services.

The company follows an agile approach to software development and focuses on delivering solutions according to project requirements. Its development process involves understanding business needs, designing suitable solutions, implementing and testing them, and providing ongoing maintenance and support. W3villa also highlights the use of modern technologies and a user-centric approach in its development work.

W3villa Technologies was founded in 2014 and has its head office in Sector 59, Noida. The company also has offices in Kanpur and the United States. During my internship, I worked with the Noida office as an Intern – Software Developer, where I was given practical assignments related to software development and computer vision data annotation.

4.3 Problem Statement

During the internship, one of the main challenges was to understand software requirements and convert them into working application features. The assigned Functional Requirement Document (FRD) contained requirements related to user management, authentication, profile management, pricing plans, payments, and administrative functions. Understanding these requirements and implementing them correctly required careful analysis and practical application of frontend, backend, and database concepts.

Another part of the internship involved working with CVAT for computer vision data annotation. This required preparing video data by creating bounding box and polygon annotations and maintaining consistency across the assigned data. The annotated work was reviewed, and errors had to be corrected when identified.

The internship therefore provided practical exposure to two different challenges in software development: converting written requirements into functional software and maintaining accuracy while preparing data for computer vision applications. Working on these tasks helped me understand the difference between learning concepts academically and applying them to actual project work.

4.4 Project Objectives

- To understand how software requirements are analysed and converted into functional application features.

- To gain practical experience in full-stack web development using the MERN stack.

- To understand and implement requirements provided through a Functional Requirement Document (FRD).

- To develop and improve features for the ORBIQ task management and productivity platform.

- To gain practical experience with frontend development, backend APIs, database operations, and application integration.

- To improve debugging and problem-solving skills by identifying and resolving issues during development and testing.
- To gain practical experience in computer vision data annotation using CVAT, including bounding box and polygon annotations.

- To understand the importance of accuracy, consistency, and review while preparing datasets for computer vision applications.

4.5 Scope of the Project

The scope of this internship included practical exposure to software development, requirement analysis, web application development, debugging, and computer vision data annotation. A major part of the development work involved understanding the Functional Requirement Document (FRD) and implementing its requirements in a task management application. This work later evolved into **ORBIQ**, a productivity-focused platform where I worked on frontend components, backend APIs, database operations, application features, and deployment-related issues.

The internship also included work with **CVAT (Computer Vision Annotation Tool)** for preparing video datasets. This involved creating bounding box and polygon annotations, getting the completed work reviewed, and making corrections wherever required. This provided practical exposure to the data preparation stage involved in computer vision projects.

Overall, the scope covered different stages of software development, from understanding requirements and building application features to testing, debugging, deployment, and data annotation. The internship was primarily focused on practical assignments and project-based learning carried out during the internship period.

4.6 Technologies and Tools Used

- Programming & Frontend Technologies: JavaScript, HTML, CSS, React.js, Vite, React Router

- Frontend Libraries & Tools: Axios, Framer Motion, Lucide React, React Hot Toast, Recharts

- Backend Development: Node.js, Express.js, REST APIs, JWT Authentication, bcrypt, Nodemailer

- Database: MongoDB, MongoDB Atlas, Mongoose

- Computer Vision & Data Annotation: CVAT (Computer Vision Annotation Tool)

- Development Environment: WebStorm 2026.2.2

- Version Control: Git and GitHub

- Deployment & Hosting: Vercel and Render

- Development Utilities: npm, dotenv, CORS, Cron Jobs

4.7 System Architecture

The system architecture of ORBIQ follows a client-server architecture in which the frontend, backend, and database work together to provide the main application functionality. The frontend is developed using React and Vite and communicates with the backend through REST APIs using Axios.

The backend is built using Node.js and Express.js. It handles user authentication, application logic, task and mission management, API requests, and communication with the database. JWT is used for authentication, while bcrypt is used for securely handling user passwords.

MongoDB is used as the main database, with MongoDB Atlas providing the cloud-based database environment. The frontend application is deployed using Vercel, while the backend is deployed using Render. This separation allows the frontend and backend to be developed, deployed, and maintained independently while communicating through the API layer.

The overall architecture used during the development of ORBIQ is represented in Figure 4.1.

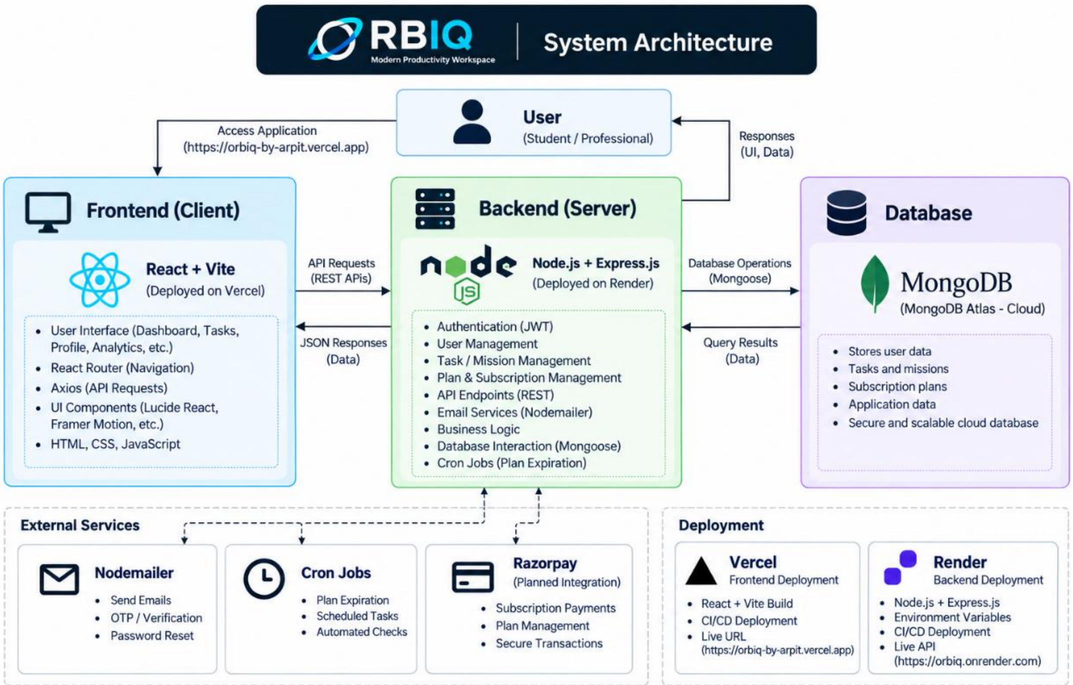

Figure 4.1: System Architecture of ORBIQ

4.8 Methodology

The internship followed a practical, task-based engineering approach where assigned work was completed according to functional requirements and reviewed during the development process. The methodology involved understanding requirements, implementing the required features, debugging issues, making corrections, and validating the completed work.

At the beginning of the internship, I was onboarded to the organization's internal communication setup using MS Teams. A technical discussion with the Tech Lead, Tarun Sir, helped assess my existing technical knowledge and understand the areas I wanted to learn during the internship. Following this, I was assigned an independent Functional Requirement Document (FRD), which served as the starting point for my software development work.

Software Development Work

The development work began with analysing the assigned FRD and understanding the features and requirements mentioned in it. The initial project was a task management application titled **TaskFlow**. Using the MERN stack (MongoDB, Express.js, React, Node.js), I worked on different parts of the application, including user authentication, REST APIs, database operations, and frontend user interface components.

As the development progressed, I expanded the original TaskFlow concept and evolved it into **ORBIQ**, a productivity-focused SaaS workspace. This transition involved redesigning the user interface, introducing productivity and analytics features, implementing subscription-related functionality and cron jobs, and working with deployment environments using Vercel and Render.

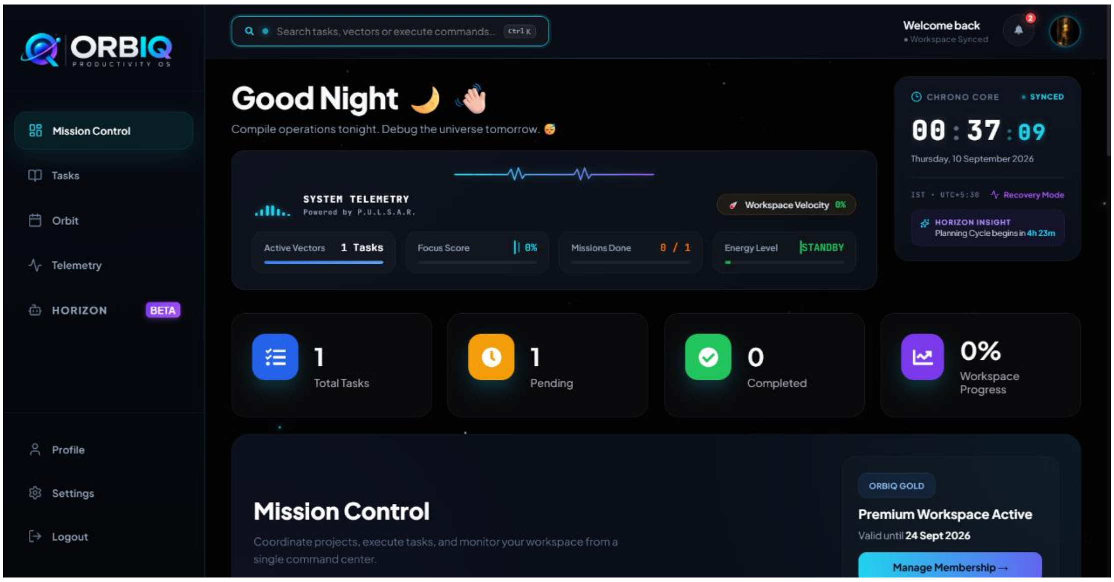

Figure 4.2: ORBIQ Mission Control Dashboard

Figure 4.2 shows the main Mission Control dashboard of **ORBIQ**. It brings together workspace metrics, system telemetry, task information, the Chrono Core, and other productivity-related elements in a single interface.

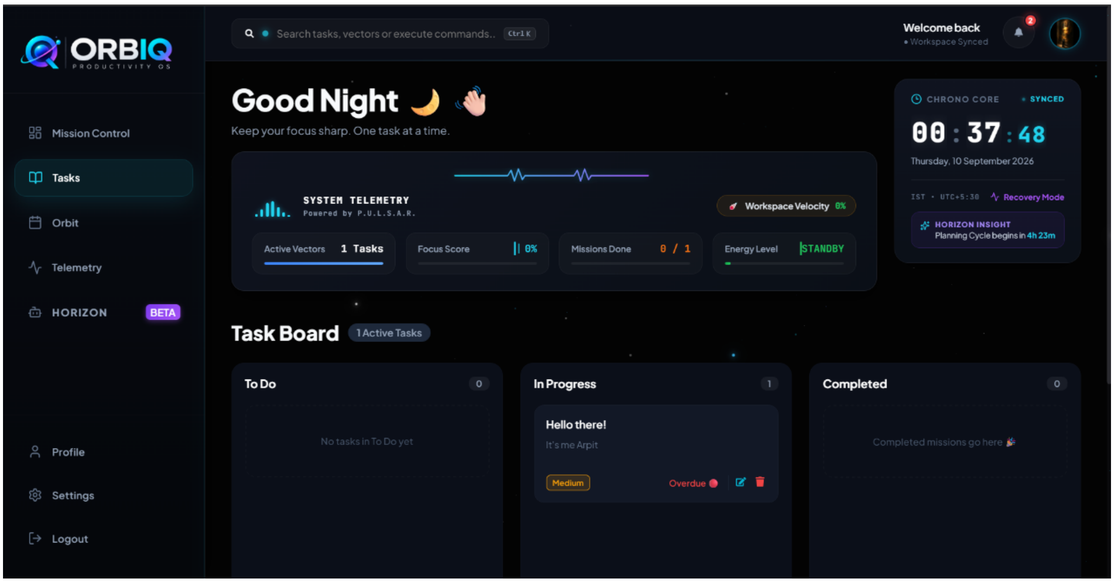

Figure 4.3: ORBIQ Task Management Interface

Figure 4.3 demonstrates the **Task Board** implemented in ORBIQ. Tasks are organized into different states such as *To Do*, *In Progress*, and *Completed*. The task cards also display information such as priority, deadlines, and available management actions.

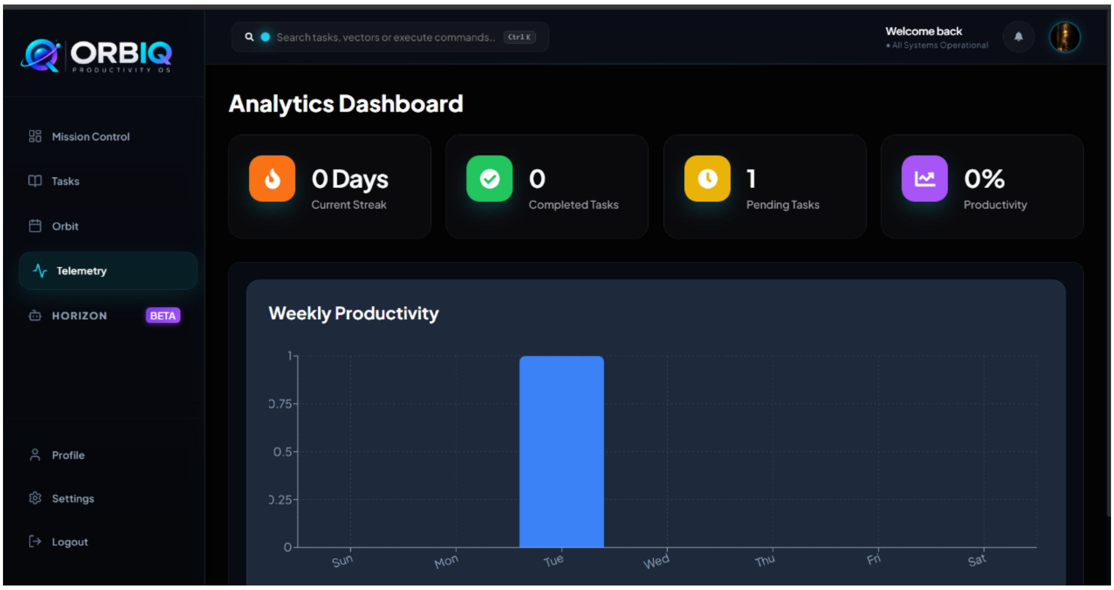

Figure 4.4: ORBIQ Analytics Dashboard

Figure 4.4 shows the **Analytics and Telemetry Dashboard** of ORBIQ. The dashboard presents application data through visual metrics and charts, including task completion, pending tasks, productivity, and weekly activity.

Computer Vision Data Annotation

In parallel with the software development work, I also worked on computer vision data annotation using **CVAT (Computer Vision Annotation Tool)**. I was provided with assigned video data and was first shown the required annotation process. After understanding the process, I worked independently on the assigned videos by creating **bounding box and polygon annotations**.

The completed annotation work was submitted for review. Whenever mistakes or issues were identified, I corrected the annotations and resubmitted the work. This process required attention to detail and consistency while working with the assigned video data.

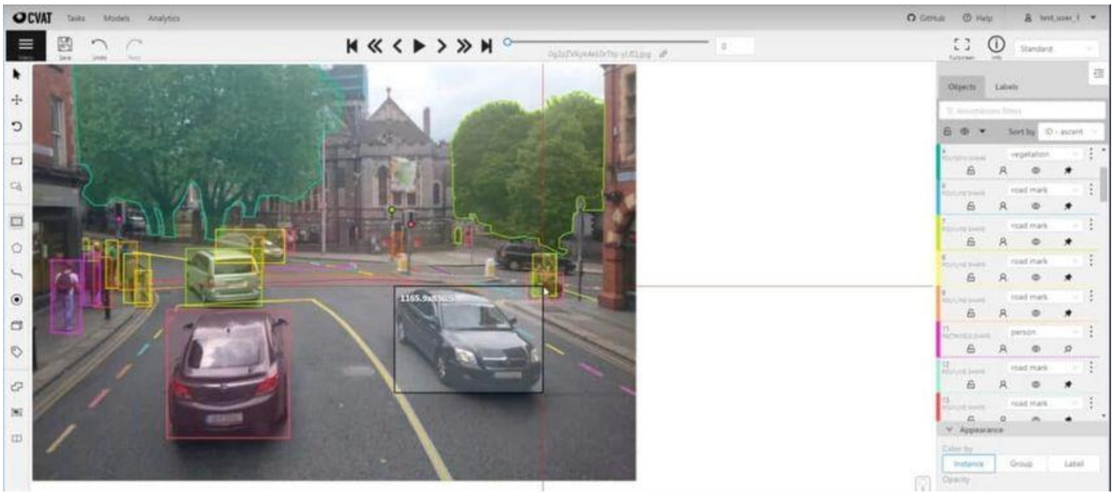

Figure 4.5: Video Data Annotation Using CVAT

Figure 4.5 illustrates the CVAT workspace used during the computer vision annotation phase. It shows annotated objects within video data along with their corresponding labels and spatial boundaries.

Overall Development Process

The overall methodology involved requirement analysis, implementation, testing, debugging, review, correction, and deployment. Working on both software development and computer vision annotation provided practical exposure to different types of technical tasks and helped me understand how development work progresses through repeated implementation and review.

4.9 Expected Outcomes

- **Practical understanding of full-stack web application development** using the MERN stack.

- Ability to **analyse Functional Requirement Documents (FRDs)** and convert requirements into functional application features.

- Improved understanding of **frontend and backend integration, REST APIs, authentication, and database operations**.

- Practical experience in **developing and improving a task management application** from the initial TaskFlow concept to the ORBIQ productivity workspace.

- Improved **debugging and problem-solving skills** through development, testing, and deployment-related issues.

- Practical experience with **computer vision data annotation using CVAT**, including bounding box and polygon annotations.

- Better understanding of the importance of **accuracy, consistency, review, and correction** in computer vision data preparation.

- Practical exposure to **version control and cloud deployment using Git, GitHub, Vercel, and Render**.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Letter issued by W3villa Technologies Private Limited is attached in this report as proof of successful completion of the internship. The original Internship Offer Letter is also included as supporting documentation. These documents confirm my internship role as an Intern – Software Developer and the internship period from 08 June 2026 to 31 July 2026.

The completion letter and offer letter have been retained in their original form to provide official proof of the internship and its associated communication.

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